RS3 — Cosmology on the deflating spacetimes: bounds on the rate of deflation, dark matter, and the Hubble tension

PROJECT spacetimes, Revision_Deflation. Supplement to both manuscripts of the revision

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0. Purpose and summary

The author's task poses two cosmological hypotheses about the superfluid of the 16-component field: that it is the dark matter, and that it resolves the Hubble tension. The supplement S3 of the main project examines both on flat extra dimensions. This supplement examines what changes when the coordinates $x5$, $x6$, $x7$ deflate.

The deflation enters cosmology in several ways, each of which requires an assumption beyond the hypotheses of the revision. The assumptions are stated in Sec. 1 and are kept apart throughout, so that the reader can see which conclusion rests on which.

Summary.

1. *If gravity is eight-dimensional* (assumption A2), Newton's constant grows at the rate of the deflation. Lunar laser ranging then bounds the present rate to gsum $< 3.24\times10^{-3}\,H_0$: over the age of the universe every coupling of the model has changed by less than a third of a per cent. *If the superfluid is the Higgs field* (the author's hypothesis H2), atomic clocks bound the rate about three thousand times more strongly still. In either case nothing that the supplements RS1 and RS4 describe along the deflation clock can have happened within cosmic history at a constant rate (Sec. 3).
2. *The Hubble tension is not relieved.* With A2 the Hubble constant inferred from the sound horizon **falls**, away from the locally measured value, by 0.2 km/s/Mpc at the largest allowed rate (Sec. 2). Without A2 the only effect of the deflation that raises it is the loss of rest mass of bound pairs, which supplies at most $+0.48$ km/s/Mpc in the examples computed (Sec. 4).
3. *Dark matter.* The estimates of the main project that involve masses only are unchanged. New is an epoch: the moment at which the pairs become bound is a scattering resonance, during which the self-interaction of the fermions exceeds the observational bound by many orders of magnitude (Sec. 6). The growth of structure changes little in amplitude and more in rate (Sec. 5).
4. *The particles that the deflation creates* (7+1) have a temperature of 2.9×10^{-33} K at the rate that lunar laser ranging allows, and below 10^{-30} K at any allowed rate: none are created today (Sec. 7).
5. *Not done.* The spectra of the microwave background, which S3 computes with its own Einstein–Boltzmann program, have not been recomputed with a time-dependent Newton's constant. The reason and the consequence are stated in Sec. 9.

0.1 How to reproduce

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation"
cargo build --release --manifest-path code/deflation-rs/Cargo.toml
./code/deflation-rs/target/release/deflation cosmology   # Secs. 2 to 4 (about a minute)
./code/deflation-rs/target/release/deflation matter      # Secs. 5 to 7
python3 code/scripts/make_figures.py hubble bounds binding
python3 code/scripts/write_doc_tables.py
python3 code/scripts/assemble_docs.py docs_src/RS3_cosmology_deflating.src.md
bash code/scripts/build_docs.sh supplement/RS3_cosmology_deflating.md
```

0.2 Symbols

symbol	in programs and data files	meaning	defined in	origin
G_N, G_8	`G_N`, `GN_early`, `GN_star`, `G_8`	Newton's constant of ordinary space, and the gravitational coupling of the eight-dimensional Einstein theory (assumption A2): $G_N = G_8/(\text{proper volume of the extra coordinates}) \propto e^{\mathrm{gsum}\, x4}$	RS3 1; M XI	new
a_{sp}, H	—	scale factor of ordinary space (assumption A1) and its rate $H = (da_{\mathrm{sp}}/dx4)/a_{\mathrm{sp}}$; H_0, h as in the main registry. (The author's Einstein-Lovelock notebooks use the letter H for the rate in the exponent of their metric; that rate corresponds here to g55, g66, g77)	RS3 1; M XI	new
t_0	`age`, `age_Gyr`	the age of the universe in the model at hand	RS3 2; M XI	new
θ_*, z_*	`theta_star`, `z_star`	the angle under which the sound horizon at last scattering is seen, $r_s(z_*)/D_A(z_*)$ with the comoving distance D_A, and the redshift z_* of last scattering at which the optical depth without reionization is 1 (the $100\,\theta_*$ of the Planck papers). The angle θ_s of the main registry is the same ratio at the maximum of the visibility function; for the model without deflation $100\,\theta_* = 1.03965$ and $100\,\theta_s = 1.040358$	RS3 1, 2; M XI	new
δ_m, f_g, σ_8	`growth`, `f_growth`, `growth_ratio`, `f_growth_ratio`, `f_sigma8_ratio`	density contrast of matter in ordinary space; its growth rate $d\ln\delta_m/d\ln a_{\mathrm{sp}}$ today; the amplitude of clustering (only ratios of σ_8 are used)	RS3 5; M XI	new
v_{EW} (main project), n_{EW}	`v_EW`, `n_EW`	the electroweak scale of the supplement S2 of the main project, and the exponent with which it would follow the coupling under the hypothesis H2: $n_{\mathrm{EW}} = d\ln v_{\mathrm{EW}}/d\ln\alpha_{VS}$	RS2 5; M XII	new
ϱ_c	`rho_c`	abbreviation in the formula for n_{EW} of the scale-invariant ladder equation: $\varrho_c = \frac{1}{2}\sqrt{\alpha_{VS}/\alpha_c - 1}$ (the letter ρ without index is a density)	RS2 5; M XII	new
m_p/m_e	`mu_pe`	the ratio of the proton mass to the electron mass (always written as this ratio in the documents; `mu_pe` in the statements of the programs)	RS2 5; RS3 3; M XI	new
G_{Fermi}, α_{em}	`G_Fermi`	the Fermi constant of the weak interaction and the fine-structure constant (they enter only in the discussion of bounds)	RS3 1, 3; M XI	new

1. The assumptions, and the equation of the expansion

The author's hypotheses fix the metric of the three deflating coordinates. Cosmology needs more. Of the following four statements, A1, A2 and HV are assumptions of this supplement, and H2 is the author's hypothesis of the main project; each is used only where stated.

A1 — ordinary space expands. The coordinates $x1, x2, x3$ carry the scale factor $a_{\text{sp}}(x4)$ of standard cosmology, with the Hubble rate $H = d\ln a_{\text{sp}}/dx4$; the hidden coordinate $x0$ is static; $x5, x6, x7$ deflate at constant rates. (The metric of DMD is the special case $a_{\text{sp}} = 1$.)

A2 — gravity is eight-dimensional. The metric obeys Einstein's equations in eight dimensions with a constant G_8. The constant of gravitation that an observer in ordinary space measures is then

$$G_N(x4) = \frac{G_8}{\text{volume of the extra coordinates}} \propto \frac{1}{V_d} \propto e^{\text{gsum}\,x4}, \qquad \frac{1}{G_N}\frac{dG_N}{dx4} = \text{gsum}. \tag{1.1}$$

Newton's constant *grows* as the extra volume shrinks: gravity was weaker in the past. (That a changing volume of extra dimensions makes the constants of four dimensions change, and that nucleosynthesis bounds the change, was pointed out in Ref. [5].)

H2 — the author's hypothesis about the Higgs field. The main project calls H2 the hypothesis that the superfluid state — the pair condensate along the hidden axis — is the Higgs field. If it holds, the electroweak scale v_{EW} is the scale of that condensate in the vacuum, which depends on the coupling α_{VS}; and α_{VS} grows with the deflation. The supplement RS2, Sec. 5, derives the consequence: the masses of the ordinary fermions follow v_{EW} and change at the rate $n_{\text{EW}}\,\text{gsum}$, with the exponent $n_{\text{EW}} = d\ln v_{\text{EW}}/d\ln\alpha_{VS}$.

HV — the vacuum value of the Vorton-Scalar sets the electroweak scale. This is a *different* assumption, not the author's: the Vorton-Scalar itself plays the role of the Higgs field. (The supplement S2 of the main project notes that the Vorton-Scalar, a singlet of all non-Abelian symmetries, cannot be the Higgs field of the Standard Model; the assumption is examined only to show what it would imply.) Under HV the Fermi constant is $G_{\text{Fermi}} = 1/(\sqrt{2}\,v_{VS}^2)$ with the four-dimensional vacuum value, and grows as e^{τ_d}, while every mass $m = y\,v/\sqrt{2}$ stays constant (RS1, Sec. 1.3).

The equation of the expansion. For a diagonal metric whose seven space-like or deflating directions have the rates H, H, H (ordinary space), 0 (the coordinate $x0$) and $-\text{g55}, -\text{g66}, -\text{g77}$, the component of Einstein's equations along the time is the sum of the products of all pairs of rates, set equal to $8\pi G_8$ times the energy density in eight dimensions. Dividing the density by the extra volume turns G_8 into G_N:

$$\boxed{3H^2 - 3\,\text{gsum}\,H + \text{gprod} = 8\pi\,G_N(x4)\,\rho} \tag{1.2}$$

with $\text{gprod} = \text{g55}\,\text{g66} + \text{g55}\,\text{g77} + \text{g66}\,\text{g77}$ and the density ρ of radiation, matter and the cosmological term in ordinary space. For $H = 0$ and $\rho \to$ the source of DMD, Sec. 1.4, this is the component ${\mathcal{G}^4}_4$ computed there. Equation (1.2) is the only one of Einstein's equations that this supplement uses; the others would determine the stresses of the unknown source that keeps the three rates constant (Sec. 9).

The calculations take three equal rates, so that $\text{gprod} = \text{gsum}^2/3$, and the densities of the model of the main project without deflation (baryons $\omega_b = 0.02237$, cold dark matter $\omega_c = 0.1200$, $H_0 = 67.36$ km/s/Mpc); the cosmological term is chosen so that Eq. (1.2) holds today. Without deflation the program reproduces the angular scale of the sound horizon and the sound horizon of the main project: $100\,\theta_* = 1.03965$ and 144.449 Mpc (check `def-C1`).

2. The Hubble constant inferred from the sound horizon

The microwave background fixes, to high precision, the angle θ_* under which the sound horizon at last scattering is seen. A model that changes the expansion history must keep that angle, and the Hubble constant that it then requires is the "inferred" Hubble constant. With A1 and A2, at fixed θ_* and fixed densities of baryons and dark matter (check `def-C2`; independently `Idef-C1`):

gsum$/H_0$	gsum (per year)	H_0 (km/s/Mpc)	change	G_N ratio	sound horizon (Mpc)	age (Gyr)
0	0	67.36	0.00	1.0000	144.45	13.81
0.00100	6.89×10^{-14}	67.30	-0.06	0.9990	144.52	13.82
0.00300	2.07×10^{-13}	67.18	-0.18	0.9971	144.66	13.84
0.00324	2.23×10^{-13}	67.16	-0.20	0.9969	144.67	13.84
0.0100	6.89×10^{-13}	66.74	-0.62	0.9905	145.14	13.89
0.0300	2.07×10^{-12}	65.48	-1.88	0.9714	146.56	14.05
0.0644	4.43×10^{-12}	63.23	-4.13	0.9384	149.12	14.34
0.100	6.89×10^{-12}	60.77	-6.59	0.9039	151.93	14.66
0.300	2.07×10^{-11}	43.41	-23.95	0.7024	172.36	17.10

(The column "G_N ratio" is Newton's constant in the early universe divided by its value today, $e^{-\mathrm{gsum}\, t_0}$ with the age t_0 of the model.) **The inferred Hubble constant falls**, with the slope $d \ln H_0 / d(\mathrm{gsum}/H_0) = -0.911$ at small rates. The reason: by Eq. (1.1) gravity was weaker in the early universe, the universe expanded more slowly at a given density, sound had more time to travel, and the sound horizon is larger; to keep the angle, the distance to last scattering must be larger, which requires a smaller Hubble constant. The deflation moves the inferred value *away* from the locally measured 73.50 ± 0.81 km/s/Mpc (the value with which the main project compares).

A remark on the term gprod*.* In Eq. (1.2) it acts as a constant addition to the left-hand side and is absorbed, almost exactly, by the cosmological term that is adjusted to today's expansion rate. The inferred Hubble constant is insensitive to it; the effect of the deflation is carried by the term linear in gsum and by $G_N(x4)$.

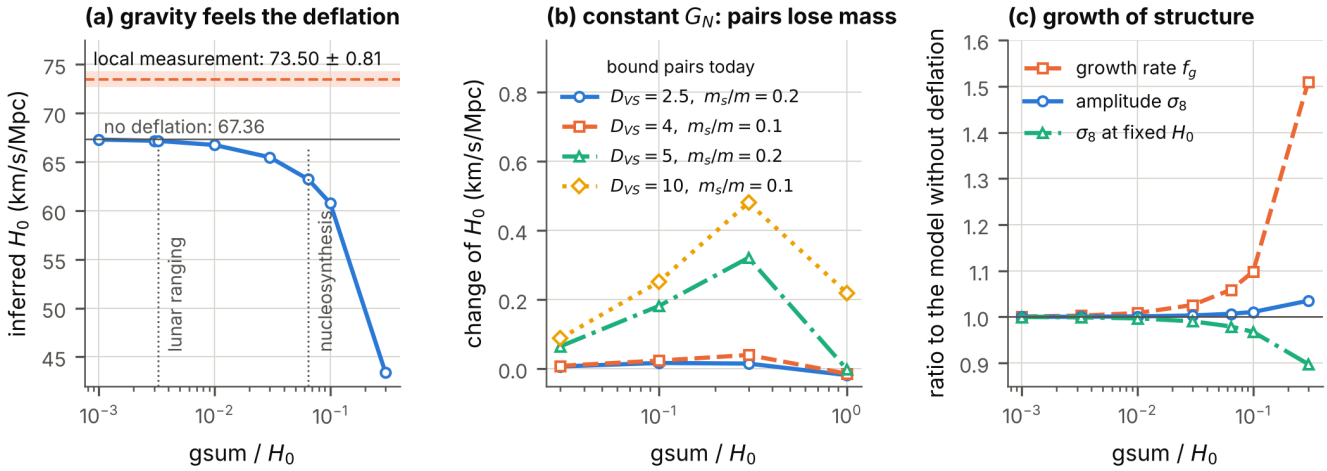


Figure 1: The Hubble constant and the growth of structure on a deflating spacetime. (a) The Hubble constant inferred from the angular scale of the sound horizon against the rate of deflation, if gravity feels the deflation (A1, A2); the bounds of Sec. 3 are marked. (b) The change of the inferred Hubble constant if Newton's constant is held fixed and bound pairs lose rest mass (Sec. 4). (c) Amplitude and rate of the growth of structure relative to the model without deflation (Sec. 5). Data: `data/hubble_deflation.csv`, `data/hubble_mass_loss.csv`, `data/growth_deflation.csv`; programs: `deflation cosmology`, `deflation matter`.

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3. How fast can the coordinates deflate today?

observation	measured	bound on gsum (per year)	bound in units of H_0	assumption
lunar laser ranging	$(dG_N/dt)/G_N$ = (7.1 ± 7.6) $\times 10^{-14}$ per year (68%)	2.230×10^{-13}	3.237×10^{-3}	gravity is eight-dimensional (A2)
primordial nucleosynthesis	G_N(nucleosynthesis)/G_N(today) = 0.99 +0.06 -0.05 (two sigma)	4.434×10^{-12}	0.0644	A2, and a rate that is constant from nucleosynthesis until today
primordial nucleosynthesis, weak interaction	G_{Fermi} at $t = 1$ s within 20 per cent (10 per cent) of today's value, fermion masses held fixed	1.615×10^{-11} (7.627×10^{-12})	0.234 (0.111)	the assumption HV: the four-dimensional vacuum value of the Vorton-Scalar sets the electroweak scale, as that of the Higgs field does (not the author's hypothesis H2, in which the Higgs field is the pair condensate); a rate that is constant from nucleosynthesis until today; no assumption about gravity
atomic clocks, mass ratio	(d mu_pe/dt)/mu_pe = (-8 ± 36) $\times 10^{-18}$ per year (68%), mu_pe = ratio of the proton mass to the electron mass	8.000×10^{-17}	1.161×10^{-6}	the author's hypothesis H2 (the superfluid state is the Higgs field), for n_EW = 1; the bound scales as 1/n_EW (data/higgs_scale_exponent.csv)
atomic clocks	$(d\alpha_{\mathrm{em}}/dt)/\alpha_{\mathrm{em}}$ = (1.0 ± 1.1) $\times 10^{-18}$ per year (68%), α_{em} = fine-structure constant	3.200×10^{-18}	4.645×10^{-8}	an assumption outside the model, which contains no electromagnetic field: stated to show how strongly such an extension would be constrained

(The bound in units of H_0 uses $H_0 = 67.36$ km/s/Mpc. Sources: lunar laser ranging, Ref. [1];

nucleosynthesis and Newton's constant, Ref. [2]; nucleosynthesis and the Fermi constant, Ref. [4]; atomic clocks, Ref. [3].)

1. **Under A2, lunar laser ranging** measures $(dG_N/dt)/G_N$ directly, and Eq. (1.1) turns its two-sigma limit into gsum $< 2.23 \times 10^{-13}$ per year, that is $3.24 \times 10^{-3}\, H_0$ (check `def-C3`). At this rate the deflation clock has advanced by gsum $t_0 = 3.1 \times 10^{-3}$ over the age t_0 of the universe.
2. **Under A2, nucleosynthesis** bounds the ratio of Newton's constant then to its value today; for a rate that was constant since, gsum $< 0.064\, H_0$. This is weaker than the first bound but does not assume that the rate today equals the rate in the past.
3. **Under HV, nucleosynthesis** bounds the change of the Fermi constant at fixed fermion masses to 10 to 20 per cent (Ref. [4]), which gives gsum < 0.111 to $0.234\, H_0$ (check `def-C5`). This bound does not involve gravity.
4. **Under the author's hypothesis H2, atomic clocks** give by far the strongest bound. The clocks compare transitions whose frequencies depend differently on the ratio m_p/m_e of the proton mass to the electron mass, and find it constant to 8.0×10^{-17} per year at two sigma (Ref. [3]). Under H2 the electron mass follows the electroweak scale and changes at the rate n_{EW} gsum, while the proton mass, which is due mostly to the strong interaction, does not. Hence
$$\text{gsum} < \frac{8.0 \times 10^{-17}\ \text{per year}}{n_{\mathrm{EW}}} = \frac{1.16 \times 10^{-6}\, H_0}{n_{\mathrm{EW}}}$$
(check `def-C6`). For $n_{\mathrm{EW}} = 1$ this is 2787 times stronger than the bound of lunar laser ranging, and it needs no assumption about gravity. The exponent n_{EW} is of order one or larger wherever it has been computed (RS2, Sec. 5).
5. **The fine-structure constant** α_{em} is bounded by the same clocks to parts in 10^{18} per year. This would bound the rate, to $4.6 \times 10^{-8}\, H_0$, only if the electromagnetic field propagated along the deflating coordinates. The model contains no electromagnetic field; the row is included to show how strongly such an extension would be constrained.
6. **In the model as it stands** — without H2, HV or an electromagnetic extension — no mass and no coupling of ordinary matter depends on the deflation, and the atomic clocks do not bound the rate at all.

Two corrections, recorded. An earlier version of this table derived a bound from the mass ratio by assuming that the electron mass follows the four-dimensional vacuum value of the Vorton-Scalar. That was an error: under HV the mass is the product of a coupling that grows and a vacuum value that falls, and is constant (provenance RP9, item R-E7). The same version attached the label H2 to the assumption HV; H2 is the author's hypothesis about the pair condensate, and its consequence is item 4 (RP9, item R-E23).

The consequence for the histories of RS1 and RS4. Under A2 the clock has advanced by at most 0.0031 during the whole of cosmic history, if the rate was constant; under H2 by at most $1.1 \times 10^{-6}/n_{\mathrm{EW}}$. The table of Sec. 6.3 shows what that means quantity by quantity; in short, *nothing happens*. A history in which pairs become bound, matter becomes abnormal or the gap grows by orders of magnitude within the age of the universe requires that gravity does not feel the deflation (A2 abandoned), that the superfluid is not the Higgs field (H2 abandoned), or that the rate was much larger in the past than it is today.

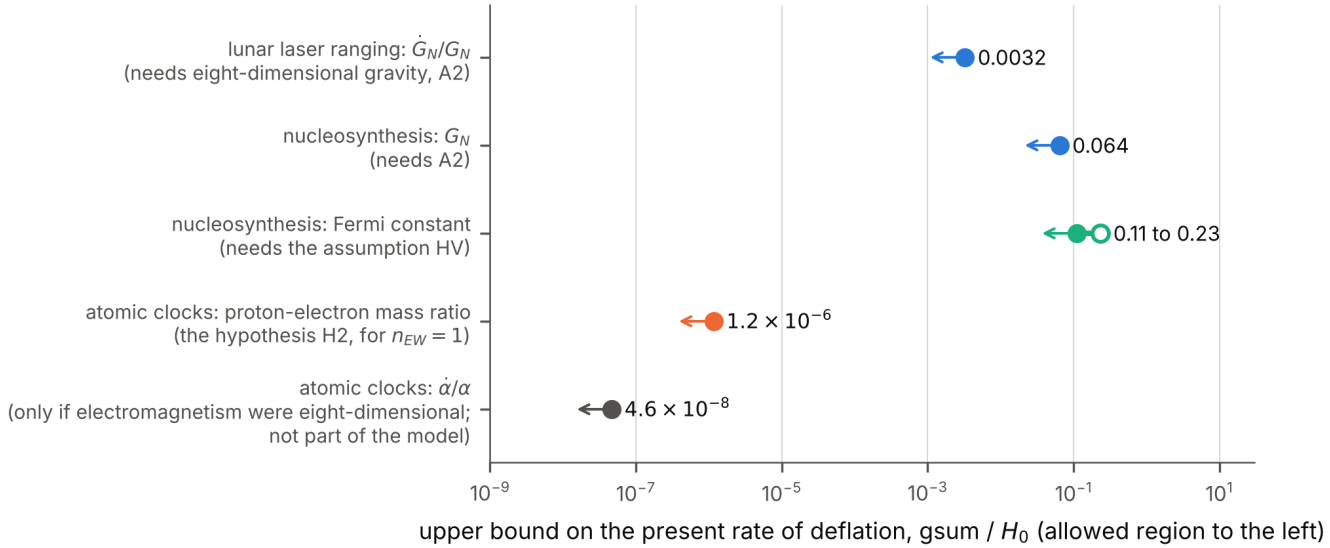


Figure 2: Upper bounds on the present total rate of deflation in units of the Hubble constant, each under the assumption stated in the table of this section. The allowed region is to the left of each bound. Data: `data/deflation_bounds.csv`; program: `deflation cosmology`.

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4. Bound pairs lose rest mass

If the dark matter consists of bound pairs of the fermions, its particles have the rest mass $2m$ minus the binding energy. The binding energy grows with the coupling (RS1, Sec. 4), so the pairs become lighter as the deflation proceeds: the density of dark matter falls slightly faster than the inverse volume. A dark matter that was heavier in the past raises the inferred Hubble constant. The program computes the effect for four examples with α_{VS} between 0.4 and 1 today, where the binding energy is at most 8.2 per cent of the rest mass (check `def-C4`):

D_{VS} today	m_s/m	α_{VS} today	gsum/H_0	gravity feels the deflation	mass today / early	H_0	change
2.50	0.20	0.50	0.00324	yes	0.99995	67.16	-0.20
2.50	0.20	0.50	0.0644	yes	0.99905	63.25	-4.11
2.50	0.20	0.50	0.0300	no	0.99953	67.37	0.01
2.50	0.20	0.50	0.100	no	0.99865	67.38	0.02
2.50	0.20	0.50	0.300	no	0.99736	67.38	0.02
2.50	0.20	0.50	1.00	no	0.99721	67.34	-0.02
5.00	0.20	1.00	0.00324	yes	0.99950	67.17	-0.19
5.00	0.20	1.00	0.0644	yes	0.99085	63.36	-4.00
5.00	0.20	1.00	0.0300	no	0.99557	67.42	0.06
5.00	0.20	1.00	0.100	no	0.98658	67.54	0.18
5.00	0.20	1.00	0.300	no	0.96882	67.68	0.32
5.00	0.20	1.00	1.00	no	0.95069	67.36	0.00
4.00	0.10	0.40	0.00324	yes	0.99994	67.16	-0.20
4.00	0.10	0.40	0.0644	yes	0.99880	63.25	-4.11
4.00	0.10	0.40	0.0300	no	0.99942	67.37	0.01
4.00	0.10	0.40	0.100	no	0.99824	67.38	0.02
4.00	0.10	0.40	0.300	no	0.99594	67.40	0.04

D_{VS} today	m_s/m	α_{VS} today	gsum$/H_0$	gravity feels the deflation	mass today / early	H_0	change
4.00	0.10	0.40	1.00	no	0.99408	67.35	-0.01
10.00	0.10	1.00	0.00324	yes	0.99933	67.17	-0.19
10.00	0.10	1.00	0.0644	yes	0.98756	63.40	-3.96
10.00	0.10	1.00	0.0300	no	0.99400	67.45	0.09
10.00	0.10	1.00	0.100	no	0.98168	67.61	0.25
10.00	0.10	1.00	0.300	no	0.95635	67.84	0.48
10.00	0.10	1.00	1.00	no	0.92342	67.58	0.22

- *If gravity feels the deflation* (rows "yes"), the inferred Hubble constant changes by between -4.11 and -0.19 km/s/Mpc at the rates that nucleosynthesis and lunar laser ranging allow: the weaker gravity of the past outweighs the loss of mass.
- *If Newton's constant is held fixed* and the deflation acts on the dark sector alone (rows "no"), rates between 0.03 and $1\,H_0$ raise the inferred Hubble constant by at most $+0.48$ km/s/Mpc, for a loss of up to 7.7 per cent of the rest mass.

The loss of rest mass is the only effect of the deflation found here that raises the inferred Hubble constant. In these examples it supplies less than a tenth of the 6.1 km/s/Mpc that separate 67.36 from the local value.

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5. The growth of structure

Under A1 and A2 the density contrast δ_m of matter in ordinary space obeys, in the Newtonian limit, the usual equation with the Newton's constant of the moment,

$$\frac{d^2\delta_m}{dx4^2} + 2H\,\frac{d\delta_m}{dx4} = 4\pi\,G_N(x4)\,\rho_m\,\delta_m .$$

The program integrates it for each model of Sec. 2 and reports the amplitude today relative to the model without deflation — the ratio of the amplitudes σ_8 for the same primordial amplitude — and the growth rate $f_g = d\ln\delta_m/d\ln a_{\rm sp}$ today (check `def-D4`):

gsum$/H_0$	H_0	σ_8 ratio	f_g ratio	$f_g\sigma_8$ ratio	σ_8 ratio, fixed H_0	f_g ratio, fixed H_0
0	67.36	1.0000	1.0000	1.0000	1.0000	1.0000
0.00100	67.30	1.0001	1.0008	1.0009	0.9997	0.9998
0.00324	67.16	1.0003	1.0027	1.0030	0.9990	0.9993
0.0100	66.74	1.0010	1.0083	1.0093	0.9968	0.9979
0.0300	65.48	1.0031	1.0258	1.0290	0.9904	0.9937
0.0644	63.23	1.0067	1.0589	1.0661	0.9792	0.9865
0.100	60.77	1.0106	1.0982	1.1098	0.9674	0.9790
0.300	43.41	1.0350	1.5100	1.5628	0.8975	0.9365

Two effects oppose each other. At a *fixed* Hubble constant, weaker gravity in the past lowers the amplitude: $d\ln\sigma_8/d(\text{gsum}/H_0) = -0.317$. But the Hubble constant that keeps the sound-horizon angle is lower, the cosmological term takes over later, and matter clusters for longer; with that Hubble constant the net change is $+0.102$ per unit of gsum$/H_0$. The amplitude therefore barely moves — $+0.03$ per cent at the bound of lunar laser ranging, $+0.67$ per cent at the bound of nucleosynthesis — while the growth rate today rises by the factors 1.0027 and 1.0589. A deflating

universe would show itself in the product $f_g\sigma_8$, which redshift-space distortions measure, rather than in σ_8.

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6. Dark matter

6.1 What does not change

The estimates of S3 that decide whether the particles of the model can be degenerate or condensed in observed halos contain the masses, the densities and the velocities only. The masses do not change along the deflation. Computed again, every number agrees with the main project: fermions with eight internal states are degenerate in the core of a dwarf galaxy only if lighter than 0.276 keV, and bound pairs are condensed there only if lighter than 262 eV (check `def-D1`). The verdict of S3 on these two possibilities stands.

6.2 Self-interaction, and the epoch of the resonance

Bound pairs. S3 requires the cross-section per mass of the pairs, $8\pi/m_s^2$ divided by the pair mass, to respect the bound of about 1 cm^2/g from colliding clusters, and derives the smallest allowed fermion mass: 918 MeV for $\alpha_{VS} = 0.1$. For a given particle the masses are constant and so is this cross-section; what the deflation changes is whether the pair *exists*: it is bound only after D_{VS} has grown to 1.67981. In terms of the coupling of the moment the smallest mass falls as $\alpha_{VS}^{-2/3}$, that is as $e^{-2\tau_d/3}$ (check `def-D2`).

The moment of binding. By RS1, Sec. 4.1, the scattering length of two unbound fermions diverges when D_{VS} passes D_c. The cross-section then rises to the limit that unitarity allows, which is larger the slower the particles. For the example particle of S3 ($m = 918$ MeV, $m_s = 54.7$ MeV) and the relative velocities 10, 156 and 1000 km/s (check `def-D3`):

- one unit of the clock before the binding the cross-section per mass is 0.84 cm^2/g at all three velocities;
- at the binding it peaks at 1.3×10^7, 5.2×10^4 and 1.3×10^3 cm^2/g;
- it exceeds 100 cm^2/g during an interval of the clock of 0.268, 0.268 and 0.258 around the binding.

A deflating universe whose dark matter consists of these fermions passes through an epoch of strong, velocity-dependent self-interaction — strongest in the smallest halos. Panel (c) of the figure in RS1, Sec. 4.1, shows the three curves. Whether the epoch lies in the observable past depends on the rate: by Sec. 3, under A2 and a constant rate it does so only if D_{VS} today exceeds D_c by less than a third of a per cent, a coincidence that nothing in the model explains.

6.3 How fast the quantities drift

With $n = d\ln(\text{quantity})/d\tau_d$, the relative change of a quantity over the age of the universe is $n\,\mathrm{gsum}\,t_0$ ($H_0 t_0 = 0.9516$ for the model without deflation; check `def-D5`):

quantity	where evaluated	n	change at the bound of lunar laser ranging	at the bound of nucleosynthesis (G_N)	at the bound of nucleosynthesis (G_{Fermi})
α_{VS}, G_{VS}, D_{VS}, C	any	1.000	3.08×10^{-3}	0.0612	0.223
g_{VS}	any	0.500	1.54×10^{-3}	0.0306	0.112
fermion mass m, mass m_s of the radial mode	any	0.000	0	0	0

quantity	where evaluated	n	change at the bound of lunar laser ranging	at the bound of nucleosynthesis (G_N)	at the bound of nucleosynthesis (G_{Fermi})
scattering length of two fermions (no bound state)	D_{VS} = 0.5	1.368	4.21×10^{-3}	0.0838	0.305
scattering length of two fermions (no bound state)	D_{VS} = 1	2.336	7.20×10^{-3}	0.143	0.521
scattering length of two fermions (no bound state)	D_{VS} = 1.5	9.092	0.0280	0.557	2.03
binding energy of the pair	D_{VS} = 1.8	30.129	0.0928	1.85	6.72
binding energy of the pair	D_{VS} = 2	12.657	0.0390	0.775	2.82
binding energy of the pair	D_{VS} = 3	4.673	0.0144	0.286	1.04
binding energy of the pair	D_{VS} = 5	3.101	9.55×10^{-3}	0.190	0.692
binding energy of the pair	D_{VS} = 10	2.456	7.57×10^{-3}	0.150	0.548
rest mass of the bound pair	D_{VS} = 5, m_s/m = 0.2	-0.162	-4.99×10^{-4}	-9.91×10^{-3}	-0.0361
rest mass of the bound pair	D_{VS} = 10, m_s/m = 0.1	-0.218	-6.73×10^{-4}	-0.0134	-0.0488
gap and critical temperature of the paired state (weak coupling)	pairing strength λ = 0.15	6.667	0.0205	0.408	1.49
gap and critical temperature of the paired state (weak coupling)	pairing strength λ = 0.2	5.000	0.0154	0.306	1.12
gap and critical temperature of the paired state (weak coupling)	pairing strength λ = 0.3	3.333	0.0103	0.204	0.744
cross-section of two fermions at low energy (Born regime, $D_{VS} \ll 1$)	D_{VS} = 0.05	2.052	6.32×10^{-3}	0.126	0.458

The couplings themselves drift with $n = 1$ or $\frac{1}{2}$. Quantities near a threshold drift much faster: the scattering length just below the binding with $n = 9.1$, the binding energy just above it with $n = 30$, the gap of a weakly paired state with $n = 1/\lambda$. These are the quantities in which a slow deflation would first be seen.

---

7. The quanta that the deflation creates

On the 7+1 spacetime the deflation creates quanta in modes that depend on a deflating coordinate, with the thermal spectrum of the temperature $T_{\mathrm{defl}} = \hbar\,\mathrm{rate}_t/2\pi k_B$ (DMD, Sec. 4.2; RS1, Sec. 2.3). For three equal rates at the largest total rates that the bounds allow (check `def-D6`):

bound	gsum$/H_0$	$\hbar\,\mathrm{rate}_t$ (eV)	T_{defl} (K)	exponent for $m = 1$ keV	exponent for $m = 1$ GeV
lunar laser ranging (A2)	0.00324	1.55×10^{-36}	2.86×10^{-33}	4.05×10^{39}	4.05×10^{45}
nucleosynthesis, G_N (A2)	0.0644	3.08×10^{-35}	5.69×10^{-32}	2.04×10^{38}	2.04×10^{44}
nucleosynthesis, G_{Fermi} (HV), 20 per cent	0.234	1.12×10^{-34}	2.07×10^{-31}	5.59×10^{37}	5.59×10^{43}
gsum $= H_0$	1.00	4.79×10^{-34}	8.85×10^{-31}	1.31×10^{37}	1.31×10^{43}

The exponent is $2\pi m/\mathrm{rate}_t$; the number of created particles of mass m per mode is $e^{-\mathrm{exponent}}$. At any rate allowed today the temperature is below 10^{-30} K and the exponent above 10^{37} for every mass above 1 keV: **no particle is created today.** Creation by deflation matters only for an epoch in which the rate was comparable with the masses, which no bound of Sec. 3 addresses.

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8. Verdicts on the two hypotheses

The Hubble tension. On the deflating spacetimes the mechanisms of the main project (S3, Sec. 3: the phase mode of the Vorton-Scalar as dark radiation) are untouched at the allowed rates, since the couplings drift by parts in a thousand; their verdict in S3 stands and is not improved. The deflation itself contributes two effects. The first, through Newton's constant, has the wrong sign and is bounded to 0.2 km/s/Mpc. The second, the loss of mass of bound pairs, has the right sign and is bounded, in the examples computed, to 0.5 km/s/Mpc. **The hypothesis that the model resolves the Hubble tension is not supported by the deflation, and under A2 it is mildly disfavoured.**

The Higgs hypothesis and the deflation together. If the superfluid is the Higgs field (H2), the atomic clocks limit the rate of deflation to $1.16 \times 10^{-6}\, H_0/n_{\mathrm{EW}}$: the two hypotheses can both hold only if the deflation is, for every purpose of this supplement, switched off today.

Dark matter. The conclusions of S3 about the mass ranges hold unchanged. The deflation adds one qualitative prediction, the resonance epoch of Sec. 6.2, and one quantitative one, the shift of the growth rate of Sec. 5. Neither is in conflict with observation at the allowed rates, and neither is detectable at the rate that lunar laser ranging allows. **The hypothesis that the superfluid is the dark matter is neither strengthened nor weakened by the deflation**; it acquires a history, whose observable part is empty if gravity is eight-dimensional and the rate constant.

---

9. What has been established, and what has not

Established, each under the assumptions named:

1. the equation of the expansion (1.2) (A1, A2) and the fall of the inferred Hubble constant, $d\ln H_0/d(\mathrm{gsum}/H_0) = -0.911$, verified by an independent integration;
2. the bounds of Sec. 3 on the present rate (under A2; under HV; under the author's hypothesis H2, from atomic clocks; under an electromagnetic extension), and that in the model as it stands the atomic clocks give none;
3. the effect of the loss of mass of bound pairs on the inferred Hubble constant (Sec. 4);
4. the growth of structure (A1, A2; Sec. 5);
5. the unchanged estimates of degeneracy and condensation, the resonance epoch, and the table of drifts (Sec. 6);
6. the temperature of the created quanta at the allowed rates (Sec. 7).

Not established.

1. *The spectra of the microwave background.* S3 of the main project computes the temperature and polarization spectra with its own Einstein–Boltzmann program and forecasts the constraints from them. The revision has **not** repeated this calculation with a time-dependent Newton's constant, for two reasons. First, the equations of the perturbations contain the stresses of the source that keeps the rates of deflation constant, which the hypotheses do not specify (item 2 below): a spectrum can be computed only under a further assumption, which is the author's to make. Second, it requires a copy of that program in which the gravitational coupling of the background and of the perturbations depends on time while the rates of Thomson scattering and of recombination do not, validated again against the constant case; it cannot be emulated by changing parameters of the existing program. What Sec. 2 establishes is the effect on the *background*. At the bound of lunar laser ranging Newton's constant at last scattering was 0.9969 of today's value; the effect on the spectra beyond the shift of the sound-horizon angle is of that relative size, and has not been computed. This item is recorded in the provenance RP0 as open, awaiting the author's instruction.
2. *The other field equations, and the theory of gravity.* Only the time component (1.2) of Einstein's equations is used. The spatial components constrain the stresses of the source that drives the deflation, which is unknown. Whether a source exists that keeps three rates constant while ordinary space expands has not been examined. DMD, Sec. 1.4a, shows that *without* expansion of ordinary space and with three equal rates the deflating spacetimes solve the equations of Einstein–Gauss–Bonnet gravity without any source. If gravity is of that kind, assumption A2 does not hold as stated: the relation (1.1) between Newton's constant and the volume of the deflating coordinates, and with it every bound and every result of this supplement that is marked A2, would have to be derived anew.
3. *A varying rate.* All bounds that reach back to nucleosynthesis assume a constant rate. A rate that was large early and is small today is neither excluded nor described.
4. *The dynamics of the resonance epoch.* Sec. 6.2 gives cross-sections; the consequences for the formation of halos during the epoch have not been simulated.
5. *Extensions.* The bound from the fine-structure constant concerns a theory with an electromagnetic field in eight dimensions, which is not the model.

Appendix A. Complete output of the programs

A.1 `deflation cosmology`

```
PROJECT spacetimes, Revision_Deflation: checks of the cosmological calculations on the deflating
 ↪  spacetime
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation cosmology
6 of 6 checks passed

[PASS] def-C1  (numerical) without deflation the background integrals of this module agree with
 ↪  the main project (its recombination calculation gives z_* = 1089.92): 100 theta_* = 1.03965
 ↪  here against 1.03965, sound horizon 144.449 Mpc against 144.449 Mpc; age 13.814 Gyr
[PASS] def-C2  (numerical) with deflation at a constant rate, gravity was weaker in the past, the
 ↪  early universe expanded more slowly at a given density, and the sound horizon is larger;
 ↪  holding 100 theta_*, omega_b and omega_c fixed, the inferred Hubble constant FALLS
 ↪  monotonically with the rate: d ln H_0/d(gsum/H_0) = -0.911 at small rates; H_0 = 66.74, 65.48,
 ↪  60.77 km/s/Mpc for gsum = 0.01, 0.03, 0.1 H_0 (67.36 without deflation). Deflation moves the
 ↪  Hubble constant away from the locally measured value
```

```
[PASS] def-C3  (numerical) bounds on the present rate of deflation if gravity is eight-dimensional
 ↪  (G_N grows at the rate gsum): lunar laser ranging, (dG_N/dt)/G_N = (7.1 +- 7.6) 10^-14 per
 ↪  year, gives gsum < 2.23e-13 per year = 3.24e-3 H_0 at two sigma; nucleosynthesis,
 ↪  G_N(nucleosynthesis)/G_N(today) > 0.94 at two sigma, gives gsum < 4.43e-12 per year = 0.064
 ↪  H_0 for a rate that was constant since then. At these rates the inferred Hubble constant
 ↪  changes by -0.199 and -4.13 km/s/Mpc: the local measurement, 73.50 +- 0.81 km/s/Mpc, cannot be
 ↪  approached. Over the age t_0 of the model without deflation (H_0 t_0 = 0.9516) the deflation
 ↪  clock has advanced by gsum t_0 = 3.1e-3 at the first of these rates
[PASS] def-C5  (numerical) bounds that do not rest on gravity: (i) under the assumption HV (the
 ↪  vacuum value of the Vorton-Scalar sets the electroweak scale) the Fermi constant grows at the
 ↪  rate gsum while every mass stays constant; nucleosynthesis with fixed masses allows G_Fermi at
 ↪  t = 1 s to differ by 10 to 20 per cent from today's value (Scherrer and Spergel 1993), which
 ↪  gives gsum < 0.111 to 0.234 H_0, weaker than the bound from Newton's constant; (ii) in the
 ↪  model as it stands (without H2 and without HV) no mass of ordinary matter depends on the
 ↪  deflation, and the measured constancy of the ratio of the proton mass to the electron mass
 ↪  does not bound the rate; (iii) the fine-structure constant would bound it, gsum < 3.2e-18 per
 ↪  year = 4.6e-8 H_0, only if the electromagnetic field propagated along the deflating
 ↪  coordinates, which is not part of the model
[PASS] def-C6  (numerical) the author's hypothesis H2 (the superfluid state is the Higgs field) on
 ↪  a deflating spacetime: the electroweak scale is then the scale of the pair condensate, which
 ↪  depends on alpha_VS, and alpha_VS grows as exp(gsum x4). With n_EW = d ln v_EW/d ln alpha_VS
 ↪  the electron mass changes at the rate n_EW gsum and the proton mass does not; atomic clocks
 ↪  (ratio of the two masses constant to 8.0e-17 per year at two sigma) then give gsum < 8.0e-17
 ↪  per year / n_EW = 1.16e-6 H_0 / n_EW, a bound 2787 times stronger than that of lunar laser
 ↪  ranging for n_EW = 1, and one that needs no assumption about gravity; over the age of the
 ↪  universe the deflation clock can then have advanced by at most 1.1e-6/n_EW. In the
 ↪  scale-invariant ladder equation of the main project n_EW = (1 + 4 rho_c^2) atan(1/(2
 ↪  rho_c))/(8 rho_c^3) + 1/(4 rho_c^2) with rho_c^2 = (alpha_VS/alpha_c - 1)/4 (agreement with
 ↪  the numerical derivative: 2.8e-8): n_EW = 15.1, 2.57, 1.15, 0.54 for alpha_VS/alpha_c = 1.25,
 ↪  2, 3, 5; it diverges at the critical coupling and exceeds 1 for alpha_VS < 3.27 alpha_c. For
 ↪  the model itself (massive fermions, massive radial mode) n_EW has not been computed
[PASS] def-C4  (numerical) dark matter of bound pairs loses rest mass while the coupling grows,
 ↪  because the binding energy increases (four examples with alpha_VS = 0.4 to 1 today; binding
 ↪  energy at most 8.2 per cent of the rest mass). If gravity feels the deflation, the inferred
 ↪  Hubble constant changes by between -4.11 and -0.19 km/s/Mpc at the rates that lunar laser
 ↪  ranging and nucleosynthesis allow: it falls, the weaker gravity of the past outweighing the
 ↪  loss of mass. If Newton's constant is held fixed and the deflation acts on the dark sector
 ↪  alone, rates between 0.03 and 1 H_0 change the inferred Hubble constant by at most +0.48
 ↪  km/s/Mpc, for a loss of up to 7.7 per cent of the rest mass (data/hubble_mass_loss.csv). The
 ↪  loss of rest mass is the only effect of the deflation that raises the inferred Hubble
 ↪  constant; in these examples it supplies less than a tenth of the 6.1 km/s/Mpc that separate
 ↪  67.36 from the local value 73.50
[PASS] ALL  (numerical) every check above passed
```

A.2 `deflation matter`

```
PROJECT spacetimes, Revision_Deflation: checks of the dark-matter estimates on the deflating
 ↪  spacetime
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation matter
6 of 6 checks passed

[PASS] def-D1  (numerical) the estimates of degeneracy and of condensation in halos contain no
 ↪  coupling: fermions with eight internal states are degenerate in the core of a dwarf galaxy
 ↪  only below 0.276 keV, bound pairs are condensed there only below 262 eV. Computed again; every
 ↪  number agrees with the file of the main project (the masses do not change along the deflation)
```

```
[PASS] def-D2  (numerical) self-interaction of bound pairs: with the coupling of today the table
 ↪  of the main project is reproduced (smallest fermion mass 918 MeV for alpha_VS = 0.1). The
 ↪  smallest mass falls as alpha_VS^(-2/3), that is, as exp(-2 tau_d/3) along the deflation. For a
 ↪  given particle the masses m and m_s are constant, hence so is the cross-section 8 pi/m_s^2 of
 ↪  the pairs; the deflation changes whether the pair exists: it is bound only after D_VS =
 ↪  alpha_VS m/m_s has grown to 1.67981
[PASS] def-D3  (numerical) the moment at which the pair becomes bound is a zero-energy resonance
 ↪  of the scattering of two fermions: the scattering length diverges at D_VS = D_c. For a
 ↪  particle with m = 918 MeV and m_s = 54.7 MeV the cross-section per mass at the relative
 ↪  velocities 10, 156 and 1000 km/s rises from 8.4e-1, 8.4e-1, 8.4e-1 cm^2/g one unit of the
 ↪  deflation clock before the binding to the peak values 1.3e7, 5.2e4, 1.3e3 cm^2/g (limit of
 ↪  unitarity at 10 km/s: 1.3e7); it exceeds 100 cm^2/g during an interval of the clock of 0.268,
 ↪  0.268, 0.258 around the moment of binding. A deflating universe in which the dark matter is
 ↪  made of these fermions passes through an epoch of strong, velocity-dependent self-interaction
[PASS] def-D4  (numerical) growth of structure when Newton's constant grows with the deflation
 ↪  (A1, A2). Two effects oppose each other. (i) Gravity was weaker in the past: at a fixed Hubble
 ↪  constant the amplitude today (sigma_8 for a given primordial amplitude) falls, d ln
 ↪  sigma_8/d(gsum/H_0) = -0.317. (ii) The Hubble constant that keeps the angular scale of the
 ↪  sound horizon is lower, the cosmological term takes over later, and matter clusters for
 ↪  longer: with that Hubble constant the net change is d ln sigma_8/d(gsum/H_0) = +0.102, a rise
 ↪  of 0.03 per cent at the rate that lunar laser ranging allows, 0.67 per cent at the rate that
 ↪  nucleosynthesis allows and 1.06 per cent for gsum = 0.1 H_0. The growth rate f_growth = d ln
 ↪  delta_m/d ln a today rises by the factors 1.0027, 1.0589, 1.0982. The amplitude of clustering
 ↪  is thus almost unchanged while its present rate of growth is not: a deflating universe would
 ↪  show itself in the product f_growth sigma_8 rather than in sigma_8 (data/growth_deflation.csv)
[PASS] def-D5  (numerical) drift of the quantities (table data/dm_drift.csv): the couplings change
 ↪  by 3.1e-3 of themselves over the age of the universe at the rate that lunar laser ranging
 ↪  allows and by 0.06 at the rate that nucleosynthesis allows; quantities near the threshold of
 ↪  binding move much faster (scattering length at D_VS = 1.5: n = 9.1; binding energy at D_VS =
 ↪  1.8: n = 30.1), and the gap of a weakly paired state moves as 1/lambda (n = 5 for lambda =
 ↪  0.2). The masses do not move
[PASS] def-D6  (numerical) the quanta that the deflation creates (7+1; modes that depend on a
 ↪  deflating coordinate): at the present rate that lunar laser ranging allows the temperature of
 ↪  the spectrum is 2.9e-33 K; at the largest rate that any of the three bounds of the
 ↪  cosmological part allows (gsum = 0.234 H_0) it is 2.1e-31 K, and hbar gsum is then 3.4e-37 of
 ↪  a mass of 1 keV. The creation of massive particles is suppressed by exp(-2 pi m/rate_t) with
 ↪  an exponent above 1e37 for every mass above 1 keV: no particles are created today. Creation
 ↪  matters only for an epoch in which the rate was comparable with the masses
[PASS] ALL  (numerical) every check above passed
```

Appendix B. Files

file	content	written by
`code/deflation-rs/src/cosmo_defl.rs`	background with deflation, sound horizon, age, growth	—
`code/deflation-rs/src/cmd_matter.rs`	dark-matter estimates, resonance, growth, drift, temperature	—
`data/hubble_deflation.csv`	Sec. 2	`deflation cosmology`
`data/deflation_bounds.csv`	Sec. 3	`deflation cosmology`
`data/hubble_mass_loss.csv`	Sec. 4	`deflation cosmology`
`data/growth_deflation.csv`	Sec. 5	`deflation matter`

file	content	written by
`data/dm_halo_estimates.csv`, `data/dm_selfinteraction.csv`, `data/dm_resonance.csv`, `data/dm_drift.csv`	Sec. 6	`deflation matter`
`data/deflation_temperature.csv`	Sec. 7	`deflation matter`
`data/cosmology_checks.txt`, `data/matter_checks.txt` (and `.csv`)	Appendix A	the two commands
`figures/fig_hubble`, `figures/fig_bounds` (`.pdf`, `.png`)	the two figures	`make_figures.py`

References

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3. R. Lange, N. Huntemann, J. M. Rahm, C. Sanner, H. Shao, B. Lipphardt, Chr. Tamm, S. Weyers and E. Peik, *Improved limits for violations of local position invariance from atomic clock comparisons*, Phys. Rev. Lett. **126**, 011102 (2021).
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6. The references of the supplement S3 of the main project, in particular its Refs. 7 (Planck Collaboration, parameters) and 20 (bounds on the self-interaction of dark matter).